**Name: Dharmit Kunal Shah** **PRN: 25070122073** **CSE A-2**

**Subject: ProgrammingwithJava**

**ASSIGNMENT No : 9**

**TITLE :**Write aJavaprogramtodemonstrate the use of the final keyword with variables, methods,and classes.

## Problem Statement

Write a Java program to demonstrate the use of the final keyword with variables, methods, and classes.

## Learning Objectives

To implement the final keyword with variables, methods, and classes.

## Theory

The final keyword restricts modification in Java programs. A final variable cannot be reassigned after initialization. A final method cannot be overridden by subclasses, while a final class cannot be inherited. This improves security, prevents accidental modifications, and increases program reliability. Final is commonly used for constants and immutable classes.

```java
final class College {
    void collegeName() {
        System.out.println(x: "College : Symbiosis Institute of Technology, Pune");
    }
}

class Person {
    final int id = 158;

    final void displayId() {
        System.out.println("ID : " + id);
    }
}

class Student extends Person {
    String name;

    Student(String name) {
        this.name = name;
    }

    void displayStudent() {
        System.out.println("Name : " + name);
    }
}

public class Main {
    Run | Debug
    public static void main(String[] args) {
        Student s = new Student(name: "Dharmit");

        s.displayStudent();
        s.displayId();

        College c = new College();
        c.collegeName();
    }
}
```

```
PS C:\Users\DHARMIT SHAH\Desktop\JAVA\Exp9> java Main.java
Name : Dharmit
ID : 158
College : Symbiosis Institute of Technology, Pune
PS C:\Users\DHARMIT SHAH\Desktop\JAVA\Exp9>
```

1. Create a **Bank Account program** where account number is final and cannot be changed.

```java
class BankAccount {
    final int accountNumber = 123456;
    String accountHolder;
    double balance;

    BankAccount(String accountHolder, double balance) {
        this.accountHolder = accountHolder;
        this.balance = balance;
    }

    void display() {
        System.out.println("Account Number : " + accountNumber);
        System.out.println("Account Holder : " + accountHolder);
        System.out.println("Balance        : " + balance);
    }
}

public class Ex1 {
    Run | Debug
    public static void main(String[] args) {
        BankAccount b = new BankAccount(accountHolder: "Dharmit", balance: 50000);
        b.display();
    }
}
```

```
PS C:\Users\DHARMIT SHAH\Desktop\JAVA\Exp9\Exercise> java Ex1.java
Account Number : 123456
Account Holder : Dharmit
Balance        : 50000.0
```

2. Create a **Library Book Management** program where the **book ISBN** is declared as **final** and cannot be changed once assigned. Display the book's ISBN, title, author, and price.

```java
class Book {
    final String isbn;
    String title;
    String author;
    double price;

    Book(String isbn, String title, String author, double price) {
        this.isbn = isbn;
        this.title = title;
        this.author = author;
        this.price = price;
    }

    void display() {
        System.out.println("ISBN   : " + isbn);
        System.out.println("Title  : " + title);
        System.out.println("Author : " + author);
        System.out.println("Price  : " + price);
    }
}

public class Ex2 {
    Run | Debug
    public static void main(String[] args) {
        Book b = new Book(isbn: "978-0134685094", title: "Generative AI", author: "Dharmit Shah", pr
        b.display();
    }
}
```

```
PS C:\Users\DHARMIT SHAH\Desktop\JAVA\Exp9\Exercise> java Ex2.java
ISBN   : 978-0134685094
Title  : Generative AI
Author : Dharmit Shah
Price  : 599.0
PS C:\Users\DHARMIT SHAH\Desktop\JAVA\Exp9\Exercise>
```